INCH-POUND

MIL-B-26994E(AL)
19 October 1992
SUPERSEDING
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13 November 1987

MILITARY SPECIFICATION

BALANCER, VEHICLE WHEEL, ON-VEHICLE TYPE

This specification is approved for use by the Armament Research, Development and Engineering Center (Tool and Equipment Engineering Division, Product Engineering/Standardization Branch), Department of the Army and is available for usage by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, electronic, vehicle wheel balancer that indicates the position and amount of running unbalance, static (up and down) vibration, and dynamic (side to side) vibration of vehicle wheels and tires while the wheels and tires are mounted on the vehicle.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-621 - Box, Wood, Nailed and Lock Corner
PPP-B-636 - Box, Shipping, Fiberboard
PPP-P-40 - Packaging and Packing of Hand Tools

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation, Methods of
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of
MIL-C-55442 - Cable Assemblies and Cord Assemblies, Packaging of

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards For Federal Services

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIl-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods
MIL-STD-1190 - Minimum Guidelines For Level C Preservation, Packing and Marking

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

US DEPARTMENT OF LABOR

29 CFR, Chapter XVII, Part 1910 - Occupational Safety and Health Standards

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATIN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11 - Plug-General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3 Wire Ground Type

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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NATIONAL ELECIRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Electrical Standards
NEMA MG1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L. Street, N.W. Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The portable electronic vehicle wheel balancer shall be new, and of the contractor's current design. The balancer shall be designed to detect running unbalance conditions in front and rear wheels of vehicles. The balancer shall be of electronic design. Detection shall be accomplished while the wheels are mounted in place on the vehicle. The complete balancer, with the applicable wheel weights, shall provide the necessary means for an operator to correct the unbalance in vehicle wheels incorporating all standard size automotive tires from 6.00 X 13 inch up to, and including 14.00 X 24 inch in both single and dual wheel applications. All parts subject to wear, breakage, or distortion shall be accessible for adjustment and repair without requiring the use of special tools or equipment.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the wheel balancer industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

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3.3.1 Reclaimed materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.4.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. When subject to removal or adjustment, these parts shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

3.4.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.4.6 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels. All units shall be supplied fully lubricated.

3.4.7 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards that can be guarded without substantial interference with operation of the balancer. All guards shall be removable to provide access to guarded parts. Except as otherwise specified by the procuring activity, the balancer shall comply with those standards promulgated under 29 CFR, Chapter XVII, Part 1910 applicable to the balancers themselves. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.3).

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3.4.8 Interchangeability. All parts of the balancer shall be manufactured to definite standards and tolerances. All respective parts of the balancer shall be interchangeable without requiring modification of the replaced part or its mating components. All spare and repair parts subject to possible future replacement shall be assigned a manufacturer's part number and shall be identified in the repair parts manual. All replacement, repair and spare parts shall provide the same or better form, fit, function, and performance as the respective parts initially installed in the balancer.

3.4.9 Surfaces. Surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

3.5 Performance and product characteristics.

3.5.1 Performance. With the tire/wheel mounted in place on the vehicle and being rotated at spinner driven speeds by means of an electric motor driven fly wheel type wheel spinner, the balancer shall be capable of detecting, within the accuracies specified herein, running balance or imbalance (unbalance), both static* (up and down) and dynamic (side to side) through the use of a vibration pickup and a synchronized strobe light circuit.

*Static in this case is meant to mean running, rotating static, which is in effect kinetic unbalance.

3.5.2 Accuracy. The accuracy of the balancer shall be such that it indicates the point of wheel unbalance within 15 degrees of the actual point of unbalance when the amount of unbalance is 1/4 ounce on a tire size of 9.50 X 16.

3.6 Details of components. The balancer shall consist of the following principal components:

a. Wheel spinner and cart assembly

(1) Motor(s)

(2) Power supply cord

b. Electronic balance indicator assembly

(1) Vibration pickup

(2) Weight meter indicator

(3) Strobe light circuit

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3.6.1 Wheel spinner and cart assembly. The spinner and cart assembly shall consist of an electric motor, fly wheel type spinner wheel, handle, controls, and a tire/wheel brake that is designed to operate against the tread surface of the tire. The wheel spinner assembly shall consist of a fly wheel type tire/wheel spinner mounted directly onto the shaft of the spinner electric motor. The shaft of the spinner motor shall operate perpendicular to the rotating plane of the tire/wheel assembly and parallel to the wheel spindle and axle shaft of the tire/wheel. The spinner shall be designed to function by positioning the fly wheel portion of the spinner against the tire tread surface that contacts the pavement. A fly wheel type wheel spinner that contacts the side wall of the tire is not acceptable. The spinner assembly shall be capable of rotating a 14.00 X 24 free turning tire, wheel, braking surface, and hub assembly from standstill to a surface speed equivalent to a vehicle velocity from 85 to 95 miles per hour (7,480 to 8,360 surface feet per minute) in 50 seconds or less. The balancer wheel/tire brake shall be capable of completely stopping wheel and tire rotation in 35 seconds or less. The spinner and cart assembly shall have wheels for portability. The spinner cart assembly shall include a means to prevent the spinner from backing away from the wheel and tire during spinning operations. A compartment shall be provided, either as a built in feature of the cart or as a separate portable container on wheels and casters, to hold at least 10 of each sized wheel weights specified in 3.6.4.4. This compartment, or stand, shall also provide for storage of the accessory equipment specified in 3.6.4 (a), (b), and (c).

3.6.2 Electronic balance indicator assembly.

3.6.2.1 Electronic design. The electronic balance indicator shall consist of a vibration pickup which detects vibration and triggers a strobe light. The strobe light flashes on and off giving the effect of a stopped wheel/tire when in fact the wheel/tire is rotating at speeds up to 90 miles per hour. The strobe light shows the location of imbalance and where to install balance correction weights to produce a balanced wheel/tire assembly. The electronic balance indicator shall also include a weight meter (or digital readout) indicator which gives an indication of the amount of imbalance.

3.6.2.1.1 Vibration pickup. The vibration pickup shall be capable of detecting the major up and down (static) and side to side (dynamic) components of vibration of the suspension system of the wheel/tire endured by imbalance of the rotating wheel/tire.

3.6.2.1.1.1 Sensitivity. The pickup shall also detect and locate through stroboscopic flash a 1/4 ounce imbalance in a 9.00 X 15 tire and wheel assembly when driven at 90 road surface miles per hour by the wheel spinner and allowed to coast down through its harmonic moments for maximum detection advantage.

3.6.2.1.1.2 Durability. The vibration pickup assembly shall be capable of withstanding a continuous vibration, acting along the same axes of the pickup sensing element, in the form of a simple harmonic motion with at least 0.030 inch amplitude (total excursion 0.060 inch) at a frequency in the range of 60 to 65 cycles per second (CPS) for a time period of at least 4 hours without visible damage or loss of accuracy as specified in 3.5.2 or loss of sensitivity as specified in 3.6.2.1.1.1.

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3.6.2.1.2 Amplifying circuit. Phase correction of the vibration pickup input signal shall be accomplished so that the indicated location for the installation of balance correction weights is at 180 degrees from the heavy spot on the vehicle wheel.

3.6.2.1.3 Meter indicator. A meter or digital readout shall indicate the relative weight required to correct balance within the accuracy limits specified in 3.5.2. The meter or digital readout shall be located to be easily readable by the operator during the balancing operation. The meter may be numbered or lettered on the dial face, and shall be marked in regular graduations.

3.6.2.1.4 Strobe light circuit. Strobe light triggering shall occur on either the positive or negative half-cycle of the signal, during the maximum excursion of the vibration pickup sensing elements. The strobe light shall be mounted with a parabolic reflector for concentrating and directing the light toward the rotating wheel. Proper triggering circuitry shall cause the lamp to flash at a rate equal to the revolutions per minute of an unbalanced rotating wheel and tire assembly, resulting in a stroboscopic effect of stopped wheel and tire motion. The strobe light shall emit light of a white or blue-white color, principally within the visible spectrum. A 1/2 inch piece of white adhesive tape mounted on the tire shall be clearly visible to an operator, when the strobe light is aimed at a distance of 30 inches from the rotating wheel in a working area illuminated to an intensity of 60 foot candles. The strobe light shall be a replaceable type with a useful life of at least 200 hours, at a rate of 15 flashes per second of continuous operation.

3.6.3 Electrical equipment.

3.6.3.1 Motors. The wheel spinner shall be powered by a single electric motor or by tandem electric motors. The motors, whether single or tandem, shall have a total power rating of 7.5 horsepower. The motor may be either single or three phase, conforming to NEMA MG1, class B insulation or better and dripproof.

3.6.3.2 Power requirements. Unless otherwise specified (see 6.2.1), the wheel spinner motor, or motors, shall operate on a voltage source of 208/230 volt, 50/60 Hertz, single phase or three phase alternating current (AC). All other circuits (amplifying, strobe light, etc.), which require a different electrical power/voltage than the main motor(s), may be transformed or powered separately (see 3.6.3.4). The amplifier and strobe light circuits shall operate on 120 volts, 50/60 Hertz, single phase AC.

3.6.3.3 Electrical components. Electrical components of the balancer, including starters, relays, switches, wiring, and their enclosures shall conform to and shall be located in accordance with NEMA Electrical Standards, except as specified herein. No component part shall be used in any application where the requirements exceed the manufacturer's rating for the part.

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3.6.3.4 Power supply cable. The balancer/wheel spinner assembly shall be furnished with a power supply cable with type SO or SJO cable cover in accordance with NFPA 70 and 29 CFR, Chapter XVII, Part 1910. Three phase motors shall be supplied with four conductor cable and single phase motors shall be supplied with three conductor cable. The power cable shall terminate with all conductors stripped and tinned for at least 1 inch. The power supply cable for the strobe light assembly, if powered separately as allowed in 3.6.3.2, shall be furnished with not less than 20 feet of three conductor flexible, electric cable with SO or SJO cable cover in accordance with NFPA 70 and 29 CFR, Chapter XVII, Part 1910, and a three prong grounding plug in accordance with ANSI C73.11. The three prong plug shall be furnished with a three to two prong outlet adapter, with a grounding lead. If the strobe light assembly is powered by an internally transformed source as allowed in 3.6.3.2, the strobe light assembly shall be connected with no less than 8 feet of electrical power cable with type SO or SJO cable cover in accordance with NFPA 70 and 29 CFR, Chapter XVII, Part 1910. The strobe light assembly shall be connected to the vibration pickup unit with no less than 8 feet of electrical power cable with type SO or SJO cable cover in accordance with NFPA 70 and 29 CFR, Chapter XVII, Part 1910.

3.6.3.5 High voltage requirement. The entire electrical system of the wheel balancer shall withstand, without damage, malfunction, breakdown, arcing, surface or air discharge, at least 2,500 volts Root Mean Square (RMS), 60 Hertz of electrical energy, at room standard ambient temperature. The electrical energy shall be applied instantaneously at full voltage of at least 2,500 volts, and shall be maintained for at least 15 seconds. Each circuit shall be isolated from other circuits and all circuits shall be isolated from electrical ground. Solid state control components may be damaged as a result of high voltage and is not cause for rejection. Damage to solid state control components may be avoided by wiring around or by-passing them, terminal to terminal. No arcing or discharge from solid state control components is acceptable.

3.6.4 Accessory equipment. The following accessory equipment shall be furnished with each balancer:

a. Wheel chock

b. Steering wheel holder

c. Wheel weight application and removal tool

d. Wheel weights

3.6.4.1 Wheel chock. A wheel chock shall be provided to be used to hold steady a wheel not being balanced.

3.6.4.2 Steering wheel holder. An adjustable height steering wheel holder shall be provided to prevent the steering wheel from turning during the operation of balancing the front wheels of the vehicles.

3.6.4.3 Wheel weight application and removal tools. Tools, for attaching and removing correction weights in a size range of one-fourth ounce to 16 ounces, shall be provided with each balancer.

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3.6.4.4 Wheel weights. Unless otherwise specified (see 6.2.1), separate boxes of 50 each 0.25, 0.50, 1.00, 1.50, 2.00, 3.00 and 4.00 ounce wheel weights for passenger cars and light trucks shall be included with each balancer. Separate boxes of 25 each 2.0, 4.0, 6.0, and 8.00 ounce wheel weights for Chrysler Wheel Part No. 4026506 shall also be included with each balancer. Separate boxes of 25 each of 2.0, 4.0, and 8.0 ounce wheel weights for heavy trucks using 14 X 24 tires shall be provided with each balancer.

3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be clean and free of all foreign matter detrimental to painting and at least one coat of primer and one coat of enamel, or equivalent, applied.

3.8 Marking. The wheel balancer shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.9 Workmanship. Workmanship of the wheel balancer, accessories and ancillary items shall be of a quality equal to that of the best commercial practices of the wheel balancer manufacturing industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When specified (see 6.2.1), first article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production units being offered for acceptance under the contract. The inspection shall consist of (a) through (f) as follows and failure of any unit to pass these inspections or tests shall be cause for rejection of the unit.

a. Product examination (see 4.5)
b. Operational accuracy test (see 4.6.1)
c. Wheel spinner test (see 4.6.2)
d. Vibration pickup endurance test (see 4.6.3)
e. Dielectric strength test (see 4.6.4)
f. Packaging inspection (see 4.7)

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or test	Lot or batch size	Sample size
Product examination (see 4.5)	1-90	100%
Operational accuracy test (see 4.6.1)	1-90	13
Wheel spinner test (see 4.6.2)	1-90	13
Vibration pickup endurance test (see 4.6.3)	1-90	13
Dielectric strength test (see 4.6.4)	1-90	13
Packaging inspection (see 4.7)	1-90	3

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100% inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

4.5 Product examination. Visually, dimensionally, and manually examine each balancer to determine conformance with the requirements of 3.2 through 3.4.9, and 3.6 through 3.9. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied to the earliest practical point in manufacturing at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the balancer and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.4.8), shall be cause for rejection.

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4.5.1 Certificate of Conformance. For purposes of quality conformance inspection only (see 4.4), and unless otherwise specified (see 6.2.1), a Certificate of Conformance shall be obtained by the contractor from the meter/indicator, strobe light, and motor suppliers certifying conformance with Section 3 and the contract.

4.6 Tests. The balancers, which have passed the examination, shall be positioned and leveled on a hard, horizontal surface, initially set up in accordance with the manufacturer's instructions, and then subjected to the following tests to determine compliance with Section 3 and the contract.

4.6.1 Operational accuracy test (9.50 X 16 inch tire. The operational tests shall be performed in accordance with manufacturer's handbook of instructions. The vehicle used in the tests shall be equipped with 9.50 X 16 tires. Using the equipment in accordance with the procedures outlined in the handbook of instructions, the operator shall test all wheels of the vehicle for running unbalance. For purposes of this test, the wheels shall be rotated by means of the spinner (see 3.6.1). The working area shall be illuminated by means of the spinner (see 3.6.1). The working area shall be illuminated to an intensity of 60-foot candles. In order to perform all balancing operations, the strobe light, when directed a distance of 30 inches from the wheel, shall provide a light of sufficient intensity to allow an operator to clearly see one-half inch, white, adhesive tape reference marker. Correction weights shall be added in the indicated amount(s) and location(s) necessary to obtain the optimum running balance. A one-fourth ounce weight shall then be attached to the outside rim of one of the front wheels. This establishes a known amount and location of running unbalance. The position of this weight shall be marked, for reference with respect to the wheel, with a one-half inch piece of white adhesive tape. The wheel shall then be balanced as accurately as the machine shall permit. Next, the original one-fourth ounce weight shall be removed and the balancer operated to determine its capability for locating and indicating weight(s) needed to correct the unbalance. The angular difference between the indicated location and the original position of the removed weight, as applied to the outside rim of the wheel, shall not exceed 15 degrees. The meter shall be capable of indicating a one-fourth ounce unbalance condition for the wheel and tire size specified in this test. Failure of the balancer to perform within specified accuracies shall be cause for rejection of the balancer (see 3.5.2).

4.6.2 Wheel spinner test. The wheel spinner shall be tested for compliance with the requirements for angular acceleration of the wheels as specified in 3.6.1. With the spinner positioned against a 14.00 X 24 inch wheel and tire, and using a stop watch for measurements of time from starting, power shall be applied to the spinner causing it to rotate the wheel up to an equivalent vehicle velocity of 90 mph in not more than 50 seconds. A series of five runs shall be made. A speed indicator with a known error of not more than one-fourth of one percent shall be used in all the tests. In all cases, both wheel and spinner shall be started from standstill and the measurement of time begun upon application of power to the spinner. After each run, the brake on the wheel spinner shall stop rotation of the wheel and tire in not more than 35 seconds. Failure of the spinner to meet the time limit for any of the five runs shall be cause for rejection of the balancer (see 3.6.1).

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4.6.3 Vibration pickup endurance test. The vibration pickup unit shall be subjected to a source of vibration acting along the same axis as the pickup sensing elements. The vibration shall be in the form of simple harmonic motion with a 0.030 inch amplitude (total excursion 0.060 inches) at a frequency in the range of 60 to 65 CPS. The duration of the complete test shall be 40 hours. Continuous operation for this period of time is not required. However, any partial run shall be not less than four hours in duration. After completion of the endurance test, the balancer shall be subjected to the accuracy test 4.6.1. Failure of the vibration pickup unit to perform satisfactorily after the complete test shall be cause for rejection of the balancer (see 3.6.2.1.1.2).

4.6.4 Dielectric strength test. A test voltage of at least 2500 volts, rms, 60 Hertz, shall be applied instantaneously between mutually insulated circuits and between insulated circuits and ground. The voltage shall be supplied by a transformer having a volt-ampere capacity of at least 500. The voltage shall be applied to an uninsulated point between each circuit and/or designation above. The voltage shall be maintained for at least 15 seconds to each circuit (see 3.6.3.5). Damage to any part or circuit evidenced by malfunction, breakdown, arcing or other surface or air discharge shall be cause for rejection. Solid state control components may be damaged as a result of high voltage and is not cause for rejection (see 3.6.3.5).

4.7 Packaging inspection. The items shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing, and marking requirements specified in Section 5.
5.

5. PACKAGING

5.1 Disassembly. The balancer may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

5.2 Preservation. Preservation shall be level A or Level C as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. Repair parts, tools, accesssories, and components that make up a complete balancer shall be cleaned in accordance with process C-1, of MIL-P-116 and dried in accordance with procedure D-1 or D-4, as applicable, of MIL-P-116.

5.2.1.2 Preservative application. Immediately following the cleaning and drying operations, all exterior, non-critical, ferrous metal surfaces of unpainted or unplated items shall be coated with Type P-1 or P-19 preservative of MIL-P-116. All interior, critical, bare, ferrous, metal surfaces shall be coated with Type P-2 preservative. Bearing surfaces (wheel or casters) shall be covered with Type P-10 or P-11 preservative, as applicable. The wheel speed indicator, disassembled from the balancer, shall be preserved in accordance with submethod IA-8 of MIL-P-116. The electric motor, also

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disassembled from the balancer, shall be preserved in accordance with MIL-E-16298. Electrical equipment with nonrotating parts, such as relays, junction boxes and amplifiers, disassembled from the balancer, shall be preserved in accordance with Level A requirements of MIL-E-17555. Hand and maintenance tools shall be preserved in accordance with Level A requirements of PPP-P-40.

5.2.1.3 Unit packed. The wheel balancer principal component parts shall be unit packed in accordance with the requirements of MIL-P-116. The following components shall be disassembled from the balancer and shall be unit protected as indicated: The wheel speed indicator shall be protected in accordance with submethod IA-8 of MIL-P-116; the electric motor shall be unit protected in accordance with MIL-E-16298; electrical equipment, such as relays, switches, junction boxes and amplifiers, shall be unit packaged in accordance with Level A requirements of MIL-E-17555; and the electrical cable and cord assemblies shall be unit packaged in accordance with Level A provisions of MIL-C-55442. Hand and maintenance tools shall be unit protected in accordance with Level A provisions of PPP-P-40. All other repair parts, accessories, and components that complete the wheel balancer, preserved as specified in 5.2.1.2, shall be unit protected using method I or III of MIL-P-116, as applicable, and shall then be individually located, one each, in a fiberboard container conforming to PPP-B-636, class weather resistant. Arrangement of the contents of the container, including cushioning, blocking, and bracing the items, shall be in accordance with the requirements of MIL-STD-1186. Closure of the container shall be in accordance with the appendix to PPP-B-636.

5.2.1.4 Consolidated container. All wrapped and all non-wrapped but boxed items that constitute one complete balancer, preserved and unit protected as specified in 5.2.1.2 and 5.2.1.3, shall be consolidated in one or more fiberboard containers, as required, conforming to PPP-B-636, class weather-resistant. Arrangement of contents, cushioning, bracing, and blocking shall be in accordance with the requirements of MIL-STD-1186. Closure of the container shall be in accordance with the appendix to PPP-B-636.

5.2.1.5 Technical publications. Publications issued with the balancer, by the contractor, shall be packaged in accordance with submethod IC-1 of MIL-P-116 and located in a protected area inside one of the fiberboard containers.

5.2.2 Level C. Level C preservation shall be as specified in MIL-STD-1190.

5.3 Packing. Packing shall be Level A, B or C, as specified (see 6.2.1).

5.3.1 Level A. Each complete balancer shall be individually packed in a shipping container conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The gross weight of the exterior container shall be governed by the weight limitations of the applicable shipping container specification. Arrangement of container contents, blocking, bracing, anchoring, and cushioning shall be in accordance with the requirements of MIL-STD-1186. Closure and strapping shall be in accordance with the requirements of the applicable specification.

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5.3.2 Level B. Level B shall be the same as Level A except shipping containers shall be domestic in lieu of overseas.

5.3.3 Level C. Level C packing shall be as specified in MIL-STD-1190.

5.4 Marking. All markings (A,B, or C) shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers portable vehicle wheel balancers used for indicating amount and position of unbalance conditions in both front and rear wheels of commercial and military-type vehicles from 6.00 X 13 inch up to, and including 14.00 X 24 inch wheel and tire sizes. The balancer shall provide the means by which an operator may correct the running unbalance with the wheels mounted on the vehicle.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. First article when required (see 3.1).

d. Special safety devices when required (see 3.4.7 and 6.3).

e. Power requirements if different (see 3.6.3.2).

f. Wheel weight requirements if different (see 3.6.4.4).

g. Painting if different (see 3.7).

h. Identification marking if different (see 3.8).

i. First article inspection if different (see 4.3).

j. Certificate of conformance if different (see 4.5.1).

k. Level of preservation and packing if different (see 5.2 and 5.3).

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6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with 29 CFR limitations and control of noise levels, radiation electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user (see 3.4.7).

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated in the contract.

6.5 Subject term (key word) listing.

Strobe light
Vibration
Wheel chock
Wheel spinner
Wheel weights

6.6 Changes fromn previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - AL

User activity:

Army - AT

Preparing activity:

Army - AL

Project No. 4910-A717

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